MSG No. 129.414 $P4'/nm'm$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-4'm'2`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,2,3,4,13,14,15,16]`
2	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[5,6,7,8,9,10,11,12]`

Table 2: Wyckoff site: `2b`, site symmetry: `-4'm'2`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[1,2,3,4,13,14,15,16]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[5,6,7,8,9,10,11,12]`

Table 3: Wyckoff site: `2c`, site symmetry: `4'm'm`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,2,7,8,9,10,15,16]`
2	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[3,4,5,6,11,12,13,14]`

Table 4: Wyckoff site: `4d`, site symmetry: `..2/m`

No.	position	mapping
1	$[0, 0, 0]$	`[1,4,5,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,3,6,7]`
3	$\left[\frac{1}{2}, 0, 0\right]$	`[9,11,13,15]`
4	$\left[0, \frac{1}{2}, 0\right]$	`[10,12,14,16]`

Table 5: Wyckoff site: `4e`, site symmetry: `..2/m`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,4,5,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,3,6,7]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[9,11,13,15]`
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[10,12,14,16]`

Table 6: Wyckoff site: `4f`, site symmetry: `2m'm'.`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,2,15,16]`
2	$\left[\frac{3}{4}, \frac{1}{4}, -z\right]$	`[3,4,13,14]`
3	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[5,6,11,12]`
4	$\left[\frac{1}{4}, \frac{3}{4}, z\right]$	`[7,8,9,10]`

Table 7: Wyckoff site: `8g`, site symmetry: `..2`

No.	position	mapping
1	$[x,\ -x,\ 0]$	`[1,4]`
2	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ 0\right]$	`[2,3]`
3	$[-x,\ x,\ 0]$	`[5,8]`
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[6,7]`
5	$\left[x+\frac{1}{2},\ x,\ 0\right]$	`[9,11]`
6	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	`[10,12]`
7	$\left[\frac{1}{2}-x,\ -x,\ 0\right]$	`[13,15]`
8	$\left[x,\ x+\frac{1}{2},\ 0\right]$	`[14,16]`

Table 8: Wyckoff site: `8h`, site symmetry: `..2`

No.	position	mapping
1	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[1,4]`
2	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[2,3]`
3	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[5,8]`
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[6,7]`
5	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}\right]$	`[9,11]`
6	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[10,12]`
7	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}\right]$	`[13,15]`
8	$\left[x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[14,16]`

Table 9: Wyckoff site: `8i`, site symmetry: `.m'.`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ z\right]$	`[1,15]`
2	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ z\right]$	`[2,16]`
3	$\left[y+\frac{1}{2},\ \frac{3}{4},\ -z\right]$	`[3,13]`
4	$\left[-y,\ \frac{3}{4},\ -z\right]$	`[4,14]`
5	$\left[\frac{3}{4},\ -y,\ -z\right]$	`[5,11]`
6	$\left[\frac{3}{4},\ y+\frac{1}{2},\ -z\right]$	`[6,12]`
7	$\left[\frac{1}{2}-y,\ \frac{1}{4},\ z\right]$	`[7,9]`
8	$\left[y,\ \frac{1}{4},\ z\right]$	`[8,10]`

Table 10: Wyckoff site: `8j`, site symmetry: `..m`

No.	position	mapping
1	$[x,\ x,\ z]$	`[1,8]`
2	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z\right]$	`[2,7]`
3	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ -z\right]$	`[3,6]`
4	$[-x,\ -x,\ -z]$	`[4,5]`
5	$\left[\frac{1}{2}-x,\ x,\ z\right]$	`[9,15]`
6	$\left[x,\ \frac{1}{2}-x,\ z\right]$	`[10,16]`
7	$\left[x+\frac{1}{2},\ -x,\ -z\right]$	`[11,13]`

continued ...

Table 10

No.	position	mapping
8	$\left[-x,\, x+\frac{1}{2},\, -z\right]$	[12,14]

Table 11: Wyckoff site: 16k, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	[2]
3	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	[3]
4	$[-y,\, -x,\, -z]$	[4]
5	$[-x,\, -y,\, -z]$	[5]
6	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	[6]
7	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	[7]
8	$[y,\, x,\, z]$	[8]
9	$\left[\frac{1}{2}-y,\, x,\, z\right]$	[9]
10	$\left[y,\, \frac{1}{2}-x,\, z\right]$	[10]
11	$\left[x+\frac{1}{2},\, -y,\, -z\right]$	[11]
12	$\left[-x,\, y+\frac{1}{2},\, -z\right]$	[12]
13	$\left[y+\frac{1}{2},\, -x,\, -z\right]$	[13]
14	$\left[-y,\, x+\frac{1}{2},\, -z\right]$	[14]
15	$\left[\frac{1}{2}-x,\, y,\, z\right]$	[15]
16	$\left[x,\, \frac{1}{2}-y,\, z\right]$	[16]